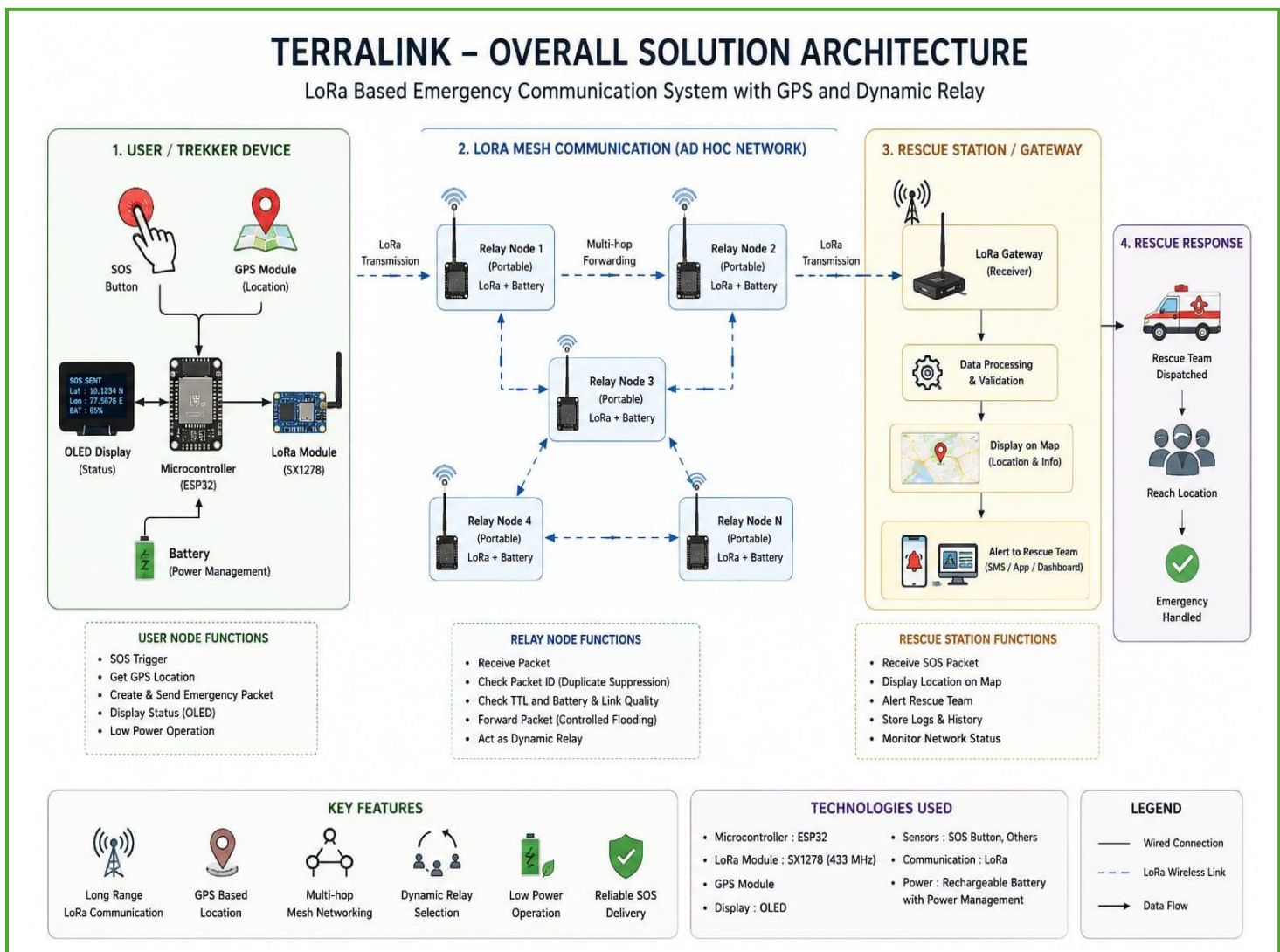


TERRALINK

SYSTEM ARCHITECTURE

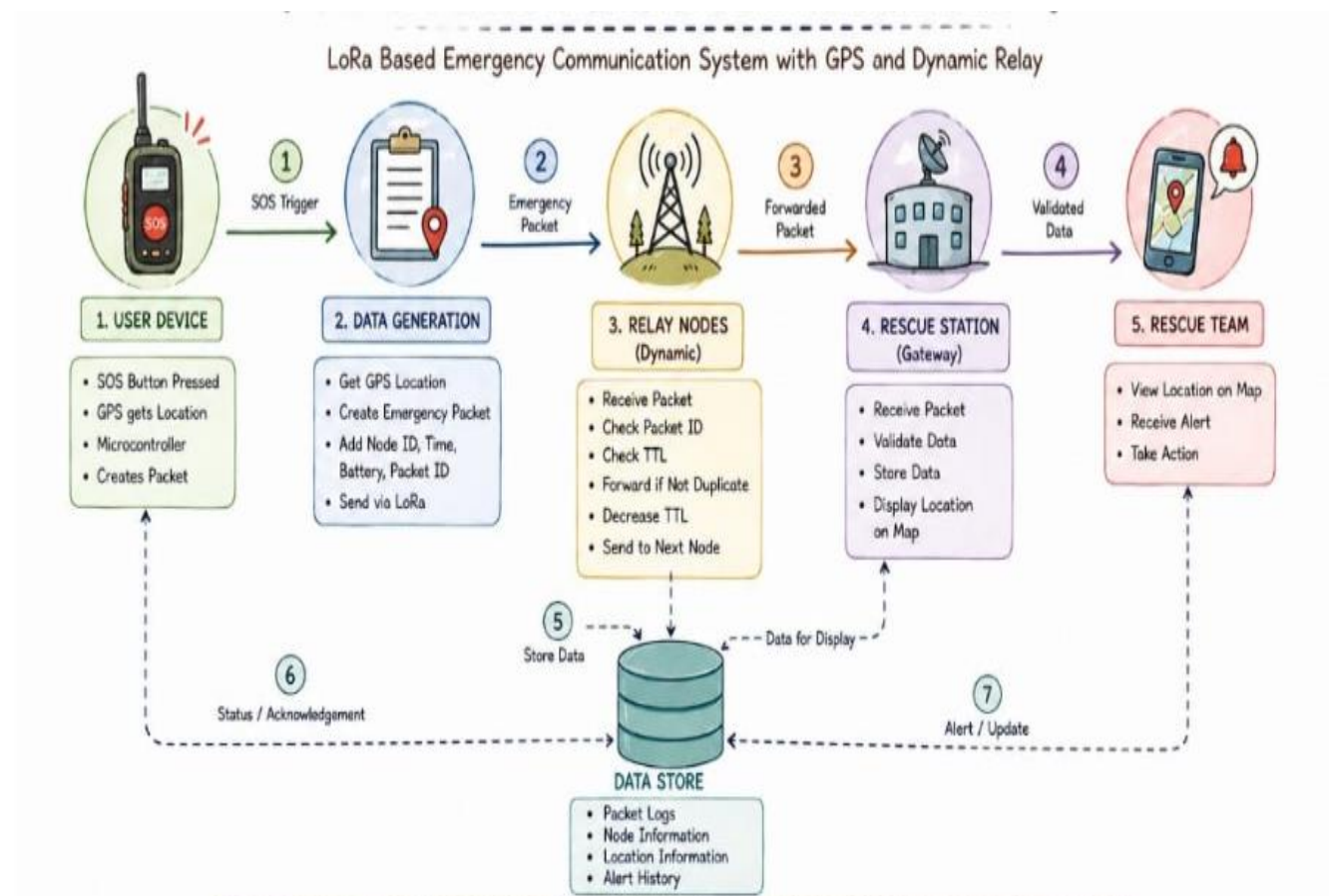
Overall Solution Architecture:

TerraLink collects the emergency trigger and GPS location data through the touch sensor, GPS module, and microcontroller. The emergency data is converted into an SOS packet and transmitted through LoRa communication. Relay nodes forward the packet until it reaches the rescue station, where the emergency message and location are received for quick response.



DATA FLOW DIAGRAM:

TerraLink collects the emergency trigger and GPS location data through the touch sensor, GPS module, and microcontroller. The emergency data is converted into an SOS packet and transmitted through LoRa communication. Relay nodes forward the packet until it reaches the rescue station, where the emergency message and location are received for quick response.



HARDWARE SOFTWARE PARTITIONING:

In TerraLink, the system is divided into hardware and software components based on their functions. The hardware includes the microcontroller, LoRa module, GPS module, touch sensor, and OLED display, which handle sensing, communication, location tracking, and data display. The software

is responsible for processing sensor inputs, collecting GPS data, creating SOS packets, controlling LoRa communication, and displaying emergency information. This partitioning ensures efficient and reliable operation of the TerraLink emergency communication system.

HARDWARE COMPONENTS			SOFTWARE MODULES		
Component	Image	Function	Module	Image / Icon	Function
Microcontroller (ESP32)		Controls all operations, processes data and manages communication.	1. Device Management		- Initialize hardware - Device ID management - Battery monitoring
LoRa Module (SX1278)		Provides long range wireless communication.	2. Sensor Management		- Read touch sensor - Detect SOS trigger - Trigger events
GPS Module (NEO-6M)		Provides location information (latitude, longitude).	3. GPS Management		- Acquire location - Parse GPS data - Provide coordinates
Touch Sensor (TTP223)		User triggers SOS alert using touch.	4. Communication (LoRa Stack)		- Create packet - LoRa transmission - Packet reception
OLED Display (0.96" I2C)		Displays status, GPS location and alerts.	5. Relay Management		- Check packet ID - TTL handling - Forward packet - Duplicate suppression
Power Supply (Li-ion Battery)		Provides power to the entire system.	6. Data Management		- Store logs - Store location - Store alerts
Antenna (433 MHz)		Enhances LoRa signal transmission.	7. User Interface		- Display status on OLED - Show location - Show alerts
			8. Power Management		- Sleep / Deep sleep modes - LoRa power control - Optimize power usage

FUNTIONAL BLOCK DIAGRAM:

Emergency Detection: Touch sensor detects an emergency.

Data Processing: Microcontroller processes the emergency signal.

Location Tracking: GPS obtains the user's location.

LoRa Transmission: Emergency message and location are transmitted through LoRa.

Relay Nodes: Relay nodes forward the message to the destination.

OLED Display: OLED displays system status and location information.

